2.45 2nd Order DEs: Modelling

Worked Solutions

1) a

Auxiliary equation:

$$4m^2 + 4m + 1 = 0 \implies (2m+1)^2 = 0 \implies m = -\tfrac{1}{2}.$$

Hence the general solution is

$$p = Ate^{-t/2} + Be^{-t/2}.$$

1) b

Substitute $p = 10$ and $t = 0$:

$$10 = A \times 0 \times e^0 + Be^0 \implies B = 10.$$

Thus $p = Ate^{-t/2} + 10e^{-t/2}$. Differentiating,

$$\frac{dp}{dt} = -\tfrac{1}{2}Ate^{-t/2} + Ae^{-t/2} - 5e^{-t/2}.$$

Substitute $p' = -2$ and $t = 0$:

$$-2 = -\tfrac{1}{2}A \times 0 \times e^0 + Ae^0 - 5e^0 \implies -2 = A - 5 \implies A = 3.$$

Therefore

$$p = 3te^{-t/2} + 10e^{-t/2}.$$

1) c

For all $t \geq 0$, $e^{-t/2} > 0$. Thus $3te^{-t/2} + 10e^{-t/2} > 0$ for all $t \geq 0$, so the modelled price never becomes negative.

1) d

From $p = 3te^{-t/2} + 10e^{-t/2}$,

$$p' = -\tfrac{3}{2}te^{-t/2} - 2e^{-t/2}, \qquad p'' = \tfrac{3}{4}te^{-t/2} - \tfrac{1}{2}e^{-t/2}.$$

The rate of decrease is greatest when $p'' = 0$:

$$(3t-2)e^{-t/2} = 0 \implies t = \tfrac{2}{3}.$$

So the rate of decrease is greatest when $t = \frac{2}{3}$ days.

1) e

One limitation is that the model assumes the price changes continuously and according to fixed coefficients throughout the time period. It does not consider promotions, competitor prices, stock levels, supplier decisions, customer demand, etc.

2) a

The auxiliary equation is $m^2 + 5m + 6 = 0 = (m+2)(m+3)$, so $m = -2$ or $m = -3$. Hence

$$\theta = Ae^{-2t} + Be^{-3t}.$$

2) b

Substitute $\theta = 0.5$ and $t = 0$:

$$0.5 = Ae^0 + Be^0 \implies A + B = 0.5. \tag{1}$$

Also $\theta' = -2Ae^{-2t} - 3Be^{-3t}$. Substitute $\theta' = 0$ and $t = 0$:

$$0 = -2Ae^0 - 3Be^0 \implies 2A + 3B = 0. \tag{2}$$

Solving (1) and (2) gives $A = \frac{3}{2}$, $B = -1$, hence

$$\theta = \tfrac{3}{2}e^{-2t} - e^{-3t}.$$

2) c

$$\theta' = -3e^{-2t} + 3e^{-3t}, \qquad \theta'' = 6e^{-2t} - 9e^{-3t}.$$

The angular speed is greatest when $\theta'' = 0$:

$$6e^t - 9 = 0 \implies e^t = \tfrac{3}{2} \implies t = \ln \tfrac{3}{2} = 0.405 \text{ s (3 SF)}.$$

2) d

$$\left.\frac{d\theta}{dt}\right|_{t=1} = -3e^{-2} + 3e^{-3} = (-3e^{-2} + 3e^{-3}) \text{ rad s}^{-1} = -0.257 \text{ rad s}^{-1} \text{ (3 SF)}.$$

2) e

As $t \to \infty$, $e^{-2t} \to 0$ and $e^{-3t} \to 0$, so $\theta(t) \to 0$. The model predicts that the door approaches the closed position, which is reasonable. However, a real door will come to rest exactly at the closed position, and stop suddenly, rather than follow a perfectly smooth exponential model for all time.

3) a

The auxiliary equation is

$$9m^2 + 6m + 2 = 0 \implies m^2 + \tfrac{2}{3}m + \tfrac{2}{9} = 0 \implies (m + \tfrac{1}{3})^2 = -\tfrac{1}{9},$$

so $m = -\frac{1}{3} \pm \frac{1}{3}i$. Hence

$$\theta = e^{-t/3}\left(A\cos\tfrac{t}{3} + B\sin\tfrac{t}{3}\right).$$

3) b

Using $\theta(0) = 0$ gives $A = 0$, so $\theta = Be^{-t/3}\sin(t/3)$. Differentiating,

$$\theta' = \tfrac{1}{3}Be^{-t/3}\cos(t/3) - \tfrac{1}{3}Be^{-t/3}\sin(t/3).$$

Since $\theta'(0) = 3$, $3 = \frac{1}{3}B$, hence $B = 9$. Therefore

$$\theta = 9e^{-t/3}\sin\tfrac{t}{3}.$$

3) c

The particular solution contains $\sin(t/3)$, which changes sign repeatedly as t increases. Therefore $\theta(t)$ alternates between positive and negative values and the carriage moves repeatedly from one side of equilibrium to the other; the motion is oscillatory. The factor $e^{-t/3}$ means the oscillations decrease in size with time.

3) d

$$\theta' = 3e^{-t/3}\cos(t/3) - 3e^{-t/3}\sin(t/3).$$

A maximum occurs when $\theta' = 0$, so $\tan(t/3) = 1$ and $t = 3\arctan 1 = 3\pi/4$. Then

$$\theta\big|_{t=3\pi/4} = 9e^{-\pi/4}\sin(\pi/4) = \frac{9\sqrt{2}}{2}e^{-\pi/4} = 2.90 \text{ rad (3 SF)}.$$

3) e

The carriage is instantaneously at rest when $\theta' = 0$. From part (d), $\tan(t/3) = 1$, with period $\pi/(1/3) = 3\pi$. The second positive time is

$$t = 3\arctan 1 + 3\pi = \frac{15\pi}{4} = 11.8 \text{ s (3 SF)}.$$

4) a

The auxiliary equation is

$$m^2 + m + \tfrac{2}{9} = 0 \implies (m + \tfrac{1}{2})^2 = \tfrac{1}{36} \implies m = -\tfrac{1}{3} \text{ or } -\tfrac{2}{3}.$$

Hence $h = Ae^{-t/3} + Be^{-2t/3}$.

4) b

From $h(0) = 0.3$, $A+B = \frac{3}{10}$. Also $h' = -\frac{1}{3}Ae^{-t/3} - \frac{2}{3}Be^{-2t/3}$ and $h'(0) = -0.3$, so $-\frac{1}{3}A - \frac{2}{3}B = -\frac{3}{10}$. Solving gives $A = -\frac{3}{10}$ and $B = \frac{3}{5}$. Thus

$$h = -\tfrac{3}{10}e^{-t/3} + \tfrac{3}{5}e^{-2t/3}.$$

4) c

At equilibrium $h = 0$:

$$-\tfrac{3}{10}e^{-t/3} + \tfrac{3}{5}e^{-2t/3} = 0 \implies -e^{-t/3} + 2e^{-2t/3} = 0 \implies -e^{t/3} + 2 = 0.$$

Thus $e^{t/3} = 2$, so $t = 3\ln 2 = 2.08$ s (3 SF).

4) d

$$h' = \tfrac{1}{10}e^{-t/3} - \tfrac{2}{5}e^{-2t/3}.$$

The greatest deviation below equilibrium occurs when $h' = 0$:

$$\tfrac{1}{10}e^{-t/3} - \tfrac{2}{5}e^{-2t/3} = 0 \implies e^{-t/3} - 4e^{-2t/3} = 0 \implies e^{t/3} - 4 = 0.$$

Thus $e^{t/3} = 4$, so $t = 3\ln 4 = 4.16$ (3 SF). At this time,

$$\begin{aligned} h &= -\tfrac{3}{10}e^{-\frac{1}{3}(3\ln 4)} + \tfrac{3}{5}e^{-\frac{2}{3}(3\ln 4)} \\ &= -\tfrac{3}{10}e^{-\ln 4} + \tfrac{3}{5}e^{-2\ln 4} \\ &= -\tfrac{3}{10}e^{\ln(4^{-1})} + \tfrac{3}{5}e^{\ln(4^{-2})} \\ &= -\tfrac{3}{10} \times \tfrac{1}{4} + \tfrac{3}{5} \times \tfrac{1}{16} = -\tfrac{3}{80} \text{ m} = -0.0375 \text{ m}. \end{aligned}$$

So the greatest deviation below equilibrium is 0.0375 m, at $t = 4.16$ s.

4) e

$$\left.\frac{dh}{dt}\right|_{t=3} = \tfrac{1}{10}e^{-1} - \tfrac{2}{5}e^{-2} = -0.0173 \text{ m s}^{-1} \text{ (3 SF)}.$$

5) a

Substituting $x = -0.02$, $x' = 0$, and $x'' = 8$ into $x'' + ax' + bx = 0$ gives $8 - 0.02b = 0$, so $b = 400$. The auxiliary equation is $m^2 + am + 400 = 0$. Repeated roots mean $a^2 - 4(1)(400) = 0$, so $a^2 = 1600$. Since $a > 0$, $a = 40$. Therefore $a = 40$ and $b = 400$.

5) b

The auxiliary equation $m^2 + 40m + 400 = 0 = (m + 20)^2$ gives $x = Ate^{-20t} + Be^{-20t}$. From $x(0) = -0.02$, $B = -0.02 = -\frac{1}{50}$. Differentiating,

$$x' = -20Ate^{-20t} + Ae^{-20t} + \tfrac{2}{5}e^{-20t}.$$

Using $x'(0) = 0$ gives $A = -\frac{2}{5}$. Thus

$$x = -\tfrac{2}{5}te^{-20t} - \tfrac{1}{50}e^{-20t}.$$

5) c

As $t \to \infty$, $e^{-20t} \to 0$ and $te^{-t} \to 0$, so $x \to 0$. The displacement never becomes positive, so the camera does not pass through equilibrium and does not oscillate. It approaches equilibrium as time increases.

6) a

For the homogeneous equation the auxiliary equation is

$$m^2 + \tfrac{3}{2}m + \tfrac{1}{2} = 0 \implies (m + \tfrac{3}{4})^2 = \tfrac{1}{16} \implies m = -\tfrac{1}{2} \text{ or } -1.$$

Thus the complementary function is $v = Ae^{-t/2} + Be^{-t}$. Try a constant particular integral $v = k$. Substitution gives $\frac{1}{2}k = 4$, so $k = 8$. Hence

$$v = Ae^{-t/2} + Be^{-t} + 8.$$

6) b

From $v(0) = 0$, $A + B = -8$. Also $v' = -\frac{1}{2}Ae^{-t/2} - Be^{-t}$ and $v'(0) = 1$, giving $-\frac{1}{2}A - B = 1$. Solving gives $A = -14$, $B = 6$, so

$$v = -14e^{-t/2} + 6e^{-t} + 8.$$

6) c

$$v' = 7e^{-t/2} - 6e^{-t}, \qquad v'' = -\tfrac{7}{2}e^{-t/2} + 6e^{-t}.$$

The greatest acceleration occurs when $v'' = 0$:

$$-\tfrac{7}{2}e^{-t/2} + 6e^{-t} = 0 \implies e^{t/2} = \tfrac{12}{7} \implies t = 2\ln\tfrac{12}{7} = 1.08 \text{ s (3 SF)}.$$

6) d

As $t \to \infty$, both exponential terms tend to zero, so $v \to 8$. The limiting speed is 8 m s^{-1}.

6) e

The model predicts that the speed approaches a constant limiting value of 8 m s^{-1}. A real car would not usually continue under exactly the same conditions. In practice, the motion would be affected by factors such as changes in driver input, road conditions, gear changes, varying resistance, running out of petrol, and the inability to travel in a straight line indefinitely due to terrain.

7) a

The homogeneous auxiliary equation is $m^2 + 3m + 2 = 0 = (m + 1)(m + 2)$, so the complementary function is $v = Ae^{-t} + Be^{-2t}$. Try $T = pt + q$, so $T' = p$ and $T'' = 0$.

$$\frac{d^2T}{dt^2} + 3\frac{dT}{dt} + 2T = 12 + t.$$

$$\begin{aligned} \frac{d^2T}{dt^2} + 3\frac{dT}{dt} + 2T &= 17 + t, \\ (0) + 3(p) + 2(pt + q) &= 17 + t, \\ (2p)t + (3p + 2q) &= t + 17. \end{aligned}$$

Comparing coefficients gives $2p = 1$, so $p = \frac{1}{2}$, and $3 \times \frac{1}{2} + 2q = 17$, so $q = \frac{31}{4}$. The particular integral is $T = \frac{1}{2}t + \frac{31}{4}$ and

$$T = Ae^{-t} + Be^{-2t} + \tfrac{1}{2}t + \tfrac{31}{4}.$$

7) b

Using $T(0) = 8$ gives $A+B = \frac{1}{4}$. Also $T' = -Ae^{-t}-2Be^{-2t}+\frac{1}{2}$ and $T'(0) = 0.25$, giving $A+2B = \frac{1}{4}$. Thus $A = \frac{1}{4}$, $B = 0$, and

$$T = \tfrac{1}{4}e^{-t} + \tfrac{1}{2}t + \tfrac{31}{4}.$$

7) c

$$T' = -\tfrac{1}{4}e^{-t} + \tfrac{1}{2}.$$

The temperature is increasing when $T' > 0$, which is equivalent to $e^t > \frac{1}{2}$. Since $t \geq 0$, $e^t \geq 1 > \frac{1}{2}$, so the temperature is always increasing.

7) d

As $t \to \infty$, the exponential term vanishes and $T \sim \frac{1}{2}t + \frac{31}{4}$. Hence $T \to \infty$.

7) e

The model may be reasonable over a short period if the heating output is steadily increasing, but it would not be realistic for a long time. The heater cannot increase output indefinitely, and the room temperature would not continue rising without limit.

8) a

The homogeneous auxiliary equation is $m^2 + m + \frac{1}{4} = 0 = (m + \frac{1}{2})^2$, giving complementary function $h = Ate^{-t/2} + Be^{-t/2}$. Try $h = pt^2 + qt + r$, so $h' = 2pt + q$ and $h'' = 2p$. Substitution gives

$$2p + (2pt + q) + \tfrac{1}{4}(pt^2 + qt + r) = t^2 + 2.$$

Comparing coefficients: $p/4 = 1$, so $p = 4$; $2p + q/4 = 0$, so $q = -32$; and $2p + q + r/4 = 2$, so $r = 104$. Therefore

$$h = Ate^{-t/2} + Be^{-t/2} + 4t^2 - 32t + 104.$$

8) b

From $h(0) = 0$, $B = -104$. Differentiating,

$$h' = -\tfrac{1}{2}Ate^{-t/2} + Ae^{-t/2} + 52e^{-t/2} + 8t - 32.$$

Using $h'(0) = 0$ gives $A + 52 - 32 = 0$, so $A = -20$. Therefore

$$h = -20te^{-t/2} - 104e^{-t/2} + 4t^2 - 32t + 104.$$

8) c

$$h' = 10te^{-t/2} + 32e^{-t/2} + 8t - 32, \qquad h'' = -5te^{-t/2} - 6e^{-t/2} + 8.$$

Thus $h''(0) = -6 + 8 = 2$. At $t = 0$, the rate of increase of the water level is increasing at 2 m s^{-2}.

8) d

As $t \to \infty$, $e^{-t/2} \to 0$ and $te^{-t/2} \to 0$, so $h \sim 4t^2 - 32t + 104$. The dominant term is $4t^2 \to \infty$, so the water level increases without bound.

8) e

The model predicts that the water level increases without bound as time increases. This is not realistic for a real tank as the tank would have a finite capacity, and the pump output could not continue increasing indefinitely. The model may be useful over a short period of time, but it would not remain valid for large values of t.

9) a

The homogeneous auxiliary equation is

$$80m^2 + 40m + 25 = 0 \implies m^2 + \tfrac{1}{2}m + \tfrac{5}{16} = 0 \implies (m + \tfrac{1}{4})^2 = -\tfrac{1}{4},$$

so $m = -\frac{1}{4} \pm \frac{1}{2}i$. The complementary function is $\theta = e^{-t/4}(A\cos(t/2) + B\sin(t/2))$. Try $\theta = p\cos(t/2) + q\sin(t/2)$. Then

$$\theta' = -\tfrac{1}{2}p\sin(t/2) + \tfrac{1}{2}q\cos(t/2), \quad \theta'' = -\tfrac{1}{4}p\cos(t/2) - \tfrac{1}{4}q\sin(t/2).$$

Substitution gives

$$(5p + 20q)\cos(t/2) + (-20p + 5q)\sin(t/2) = 136\cos(t/2),$$

so $5p + 20q = 136$ and $-20p + 5q = 0$. Thus $p = \frac{8}{5}$, $q = \frac{32}{5}$, giving the general solution

$$\theta = e^{-t/4}(A\cos(t/2) + B\sin(t/2)) + \tfrac{8}{5}\cos(t/2) + \tfrac{32}{5}\sin(t/2).$$

Using $\theta(0) = 0.4$ gives $A = -\frac{6}{5}$. Differentiating the general solution and using $\theta'(0) = -0.5$ gives $B = -8$. Therefore

$$\theta = e^{-t/4}(-\tfrac{6}{5}\cos(t/2) - 8\sin(t/2)) + \tfrac{8}{5}\cos(t/2) + \tfrac{32}{5}\sin(t/2).$$

9) b

As $t \to \infty$, $e^{-t/4} \to 0$, so for large t,

$$\theta \sim \tfrac{8}{5}\cos(t/2) + \tfrac{32}{5}\sin(t/2).$$

The platform approaches a steady oscillation with period $2\pi/(1/2) = 4\pi$.

9) c

Write $\frac{8}{5}\cos(t/2) + \frac{32}{5}\sin(t/2) \equiv R\cos(t/2 - \alpha)$. Comparing coefficients gives $R\cos\alpha = \frac{8}{5}$ and $R\sin\alpha = \frac{32}{5}$. Hence

$$R = \sqrt{(\tfrac{8}{5})^2 + (\tfrac{32}{5})^2} = \tfrac{8\sqrt{17}}{5}, \qquad \tan\alpha = 4, \quad \alpha = \arctan 4 = 1.33 \text{ (3 SF)}.$$

Therefore the steady-state motion is

$$\frac{8\sqrt{17}}{5}\cos\left(\tfrac{1}{2}t - \arctan 4\right).$$

9) d

$$R = \frac{8\sqrt{17}}{5} = 6.596969\ldots > 6.5.$$

The steady-state amplitude exceeds 6.5, so the platform cannot be regarded as safe.

10) a

The homogeneous auxiliary equation is $36m^2+108m+81 = 0$, or $(m+\frac{3}{2})^2 = 0$. The complementary function is $T = Ate^{-3t/2} + Be^{-3t/2}$. Try $T = pt^2e^{-3t/2} + qe^{t/6}$. Then

$$T' = -\tfrac{3}{2}pt^2e^{-3t/2} + 2pte^{-3t/2} + \tfrac{1}{6}qe^{t/6},$$
$$T'' = \tfrac{9}{4}pt^2e^{-3t/2} - 6pte^{-3t/2} + 2pe^{-3t/2} + \tfrac{1}{36}qe^{t/6}.$$

Substitution into $36T'' + 108T' + 81T = 216e^{-3t/2} + 100e^{t/6}$ gives

$$(72p)e^{-3t/2} + (q + 18q + 81q)e^{t/6} = 216e^{-3t/2} + 100e^{t/6}.$$

Thus $72p = 216$ and $100q = 100$, so $p = 3$, $q = 1$. The general solution is

$$T = Ate^{-3t/2} + Be^{-3t/2} + 3t^2e^{-3t/2} + e^{t/6}.$$

10) b

Using $T(0) = 4$ gives $B = 3$. Then

$$T' = -\tfrac{3}{2}Ate^{-3t/2} + Ae^{-3t/2} - \tfrac{9}{2}e^{-3t/2} - \tfrac{9}{2}t^2e^{-3t/2} + 6te^{-3t/2} + \tfrac{1}{6}e^{t/6}.$$

Using $T'(0) = -\frac{1}{3}$ gives $-\frac{1}{3} = A - \frac{9}{2} + \frac{1}{6}$, so $A = 4$. Hence

$$T = 4te^{-3t/2} + 3e^{-3t/2} + 3t^2e^{-3t/2} + e^{t/6} = (3t^2 + 4t + 3)e^{-3t/2} + e^{t/6}.$$

10) c

As $t \to \infty$, $(3t^2 + 4t + 3)e^{-3t/2} \to 0$ because the exponential tends to zero faster than the quadratic increases, while $e^{t/6} \to \infty$. Thus $T \sim e^{t/6}$ and the temperature increases without bound.

10) d

$$T(10) = (3 \times 10^2 + 4 \times 10 + 3)e^{-3(10)/2} + e^{10/6} = 0.0001049\ldots + 5.29449\ldots = 5.29°\text{C (3 SF)}.$$

10) e

Set $20 = (3t^2 + 4t + 3)e^{-3t/2} + e^{t/6}$. By the time the temperature is 20°C, $(3t^2 + 4t + 3)e^{-3t/2} \approx 0$. Estimate by solving

$$20 = e^{t/6} \implies t = 6\ln 20 = 17.97439\ldots = 18.0 \text{ minutes (3 SF)}.$$